

DR. ZAINAB KHAN

Environmental Geographer | Geospatial & Remote Sensing Researcher | Climate Change & Natural Hazards

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PROFESSIONAL PROFILE

Environmental Geographer and Geospatial Researcher with a PhD in Geography and postdoctoral research experience at Aligarh Muslim University and IIT Roorkee. My research focuses broadly on **climate change, natural hazards and spatial/geospatial analysis**, using **GIS, remote sensing, spatial modelling, artificial intelligence, machine learning and deep learning**. My work has addressed landslide susceptibility, Himalayan river-basin risk, forest-fire vulnerability, land-surface-temperature change, groundwater mapping, carbon sequestration, lake ecosystems, phytoplankton ecology and climate-sensitive Himalayan landscapes. Environmental monitoring, sensors, UAV observations and citizen-generated information are used where relevant within this broader geospatial framework. I also develop research-oriented geospatial platforms, including **HazWatch** and **WeatherScope**.

CORE RESEARCH INTERESTS

- Climate change and environmental change
- Natural hazards and multi-hazard assessment
- Landslides, flash floods, floods and GLOFs
- GIS, remote sensing and spatial analysis
- Geospatial artificial intelligence
- Machine learning and predictive modelling
- Deep learning applications in geospatial research
- Spatial statistics and environmental modelling
- Himalayan environmental systems
- Earth observation and image processing
- Environmental and spatial data integration
- Citizen science and geospatial application development

PROFESSIONAL & RESEARCH EXPERIENCE

Postdoctoral Researcher - Geospatial Engineering Group

Indian Institute of Technology Roorkee (IIT Roorkee), India | 6-month appointment

- Worked on an AI-enabled agricultural monitoring framework integrating satellite, UAV/drone and PhenoCam observations.
- Applied geospatial and Earth-observation approaches to crop-health monitoring, stress detection and spatial analysis.
- Contributed to workflows integrating remotely sensed and field-scale information.

ICSSR Postdoctoral Fellow - Department of Geography

Aligarh Muslim University, Aligarh, India | 2023-2025

- **Research project:** "Impact Assessment of Climate Change and Anthropogenic Activities on Landslides in Nainital District of Uttarakhand."
- Investigated climatic, topographic, land-cover and anthropogenic controls on landslide occurrence.
- Applied GIS, remote sensing, statistical analysis and machine-learning approaches to landslide research.
- Collaborated internationally on landslide research with Prof. Daisuke Komori, Japan.

Research Assistant - Department of Geography

Aligarh Muslim University, Aligarh, India | 2023

- Handled the **GIS, remote sensing and data-analysis components** of the research work.
- Supported spatial data processing, geospatial analysis and remote-sensing workflows.

Founder & Developer - HazWatch

Citizen-science multi-hazard geospatial platform for India | 2026-Present

- Conceptualised and developed a citizen-powered multi-hazard reporting and geospatial platform for India.
- Supports geolocated reporting of landslides, floods, flash floods, forest fires, avalanches, GLOFs, debris/mud flows, droughts, heat waves and cloudbursts.
- Integrates photographic evidence, incident metadata, spatial mapping, moderation and anonymised research-data access.
- Technology stack includes React Native/Expo, FastAPI, MongoDB, Firebase and Leaflet/OpenStreetMap.

Developer - WeatherScope

Interactive global weather and atmospheric web application

Map-based weather explorer integrating current weather, annual patterns, wind, air quality, solar radiation, elevation and astronomical information.

EDUCATION

Ph.D. in Geography

2023

Aligarh Muslim University, Aligarh, India

Thesis: "Environmental Impact Assessment of Climate Change in Kumaun Region-Uttarakhand."

Postgraduate Diploma in Remote Sensing & GIS

2017

Jamia Millia Islamia, New Delhi

Image processing, image classification, ArcGIS and ERDAS Imagine; passed with distinction.

M.A. in Geography

2016

Jamia Millia Islamia, New Delhi

Economic Geography, Statistics, Physical Geography, Remote Sensing and GIS; passed with distinction.

B.A. (Hons) Geography

2014

Kamala Nehru College, University of Delhi

First Division.

Certificate in French Language

2013

Kamala Nehru College, University of Delhi

FELLOWSHIPS & ACADEMIC RECOGNITION

2018	Maulana Azad National Fellowship	National fellowship supporting doctoral research.
2023	ICSSR Postdoctoral Fellowship	Indian Council of Social Science Research postdoctoral fellowship.
2025	Chanakya Postdoctoral Fellowship	IIT Tirupati.
2018 & 2019	UGC-NET	Qualified twice: January 2018 and June 2019.

SELECTED RESEARCH & METHODOLOGICAL EXPERIENCE

Climate Change Analysis - Kumaun Himalaya

- Analysed multi-decadal precipitation, snow-cover and land-surface-temperature trends.
- Used NDSI for snow-cover analysis and remote-sensing approaches for LST estimation.
- Applied interpolation and spatial-analysis techniques in ArcGIS.

Lake Ecology & Spatial Modelling

- Conducted geocoded field measurements of water temperature, pH, dissolved oxygen and electrical conductivity.
- Collected Secchi-depth and phytoplankton observations.
- Applied spatial interpolation, PCA, Shannon entropy, 3D visualisation and species-distribution modelling.
- Developed continuous probability surfaces using Random Forest and environmental conditioning variables.

Hydrological & Hazard Modelling

- Experience with HEC-RAS for GLOF and flash-flood modelling.
- Developed flood hydrographs and worked with 2D/3D hydrological modelling.
- Used microwave remote sensing for land subsidence, glacier movement and hazard-related applications.

TECHNICAL SKILLS

GIS & Spatial Analysis	ArcGIS, QGIS, Google Earth, georeferencing, overlay analysis, site suitability, interpolation, raster analysis, DEM, slope/aspect analysis, area/volume estimation and cartography.
Remote Sensing	ERDAS Imagine, SNAP, Google Earth Engine, stacking, filtering, mosaicking, supervised/unsupervised classification, segmentation and microwave remote sensing.
Programming & Computational Methods	Python, R, JavaScript, RStudio, Jupyter and PyCharm.
AI & Machine Learning	Random Forest, Support Vector Machine, Boosted Regression Tree, Artificial Neural Networks, GLM, GAM, predictive mapping and species-distribution modelling.
Statistics	ANOVA, ANCOVA, regression, PCA, cluster analysis, canonical correlation, hypothesis testing, chi-square, t-test, z-test, Shannon entropy and quantitative SWOT.
Indices & Environmental Products	NDVI, NDBI, NDWI, NDGI, NDSI, SAVI, MSAVI2, LST, insolation and terrain derivatives.
Hydrological Modelling	HEC-RAS, GLOF modelling, flash-flood modelling and hydrograph development.

COMPLETE PUBLICATIONS & RESEARCH OUTPUTS

Research outputs are grouped by type so that journal articles, book chapters, preprints and conference papers are clearly distinguished.

Peer-reviewed Journal Articles

- 1. Predicting Forest Fire Risk and Carbon Pool Vulnerability in the Himalayas: A Machine Learning Approach**
Earth Systems and Environment, 10(4), 4387-4413 | 2026 (online 2025) | First author | DOI: 10.1007/s41748-025-00863-0
- 2. Analysing the Physicochemical Dynamics and Biodiversity Patterns in Bhimtal, Nainital, and Sariyatal Lakes**
Discover Water, 5, Article 76 | 2025 | First author | DOI: 10.1007/s43832-025-00270-6
- 3. Mapping Landslide Susceptibility and Risk Assessment on Fragile Ecosystem of Himalayan River Basins**
All Earth, 37(1), Article 2490326 | 2025 | First author | DOI: 10.1080/27669645.2025.2490326
- 4. Groundwater Contamination, Fate, and Transport of Fluoride and Nitrate in Western Jilin, China: Implications for Water Quality and Health Risks**
Environmental Chemistry and Ecotoxicology, 7, 1189-1202 | 2025 | Co-author | DOI: 10.1016/j.enceco.2025.04.010
- 5. Temporal Trends and Future Projections: Analysing Land Surface Temperature in the Kumaun Himalayas Using Spatial Time Series Analysis**
Remote Sensing Applications: Society and Environment, 37, 101426 | 2025 | First author | DOI: 10.1016/j.rsase.2024.101426

6. Analysing the Potential Impacts of Land Use Land Cover (LULC) Transformation on Present and Future Carbon Sequestration Capabilities in the Central Himalayas

Discover Geoscience, 2, Article 92 | 2024 | First author | DOI: 10.1007/s44288-024-00097-z

7. Integrating Gravity Models and Network Analysis in Logistical Strategic Planning: A Case of the India Middle-East Europe Economic Corridor (IMEC)

Maritime Economics & Logistics, 27(1), 147-182 | 2024 | Co-author | DOI: 10.1057/s41278-024-00289-x

8. Comparing the Performance of Machine Learning Algorithms for Groundwater Mapping in Delhi

Journal of the Indian Society of Remote Sensing | 2024 | First author | DOI: 10.1007/s12524-023-01789-8

9. Aggregating an Economic Model and GIS to Explore Trade Potentials of India-Caspian Countries and a Way Forward for INSTC

Research in Globalization, 7, 100154 | 2023 | First author | DOI: 10.1016/j.resglo.2023.100154

10. Hydro-geochemical Assessment of Ground Water for Drinking and Agricultural Purposes and Potential Human Health Risk in Aligarh City, India

Chemical Engineering Journal Advances, 16, 100547 | 2023 | Co-author | DOI: 10.1016/j.ceja.2023.100547

11. Estimating Photosynthetically Active Euphotic Layer in Major Lakes of Kumaun Region Using Secchi Depth Water, Air, & Soil Pollution, 234(9), 597 | 2023 | First author | DOI: 10.1007/s11270-023-06612-1

12. Analysing Land Use Land Cover Changes and Urban Built-up Growth Using NDBI and Shannon's Entropy Model: A Spatio-Temporal Analysis of Bulandshahr City, India

Asian Profile, 50(4), 361-380 | 2022 | Co-author

13. A District-Level Vulnerability Assessment of Next COVID-19 Variant (Omicron BA.2) in Uttarakhand Using Quantitative SWOT Analysis

Environment, Development and Sustainability | 2022 | First author | DOI: 10.1007/s10668-022-02727-3

14. Predicting the Effects of Climate Change on Prospective Banj Oak (Quercus leucotrichophora) Dispersal in Kumaun Region of Uttarakhand Using Machine Learning Algorithms

Modeling Earth Systems and Environment | 2023 (online 2022) | First author | DOI: 10.1007/s40808-022-01485-5

15. Monitoring and Assessment of Physico-chemical Properties of Groundwater in South Delhi (Shaheen Bagh) Using Inferential Statistics

Adalya Journal, 11(1) | 2022 | First author | DOI: 10.37896/aj11.1/006

16. Modelling Physicochemical Properties of Bhimtal Lake Using Geospatial Techniques

International Journal of All Research Education and Scientific Methods (IJARESM), 9(2) | 2021 | First author

17. Modelling Correlation Between Nightlights and Population Using Defense Meteorological Satellite Program (DMSP) Data Through Statistical Inferences

Journal article | 2021 | First author

Book Chapters

18. Aggregating Machine Learning and GIS for Flood Susceptibility Modeling on Agricultural Land in Banda District, Uttar Pradesh, India

Climate Change, Resilience and Ecosystem Services, Springer Nature, pp. 291-310 | 2026 | Co-author | DOI: 10.1007/978-3-031-96977-5_17

19. GIS-Based Micro-Solar Power Plant Site Selection in Sundarban Delta of India: A Siting Model for Sustainable Utility of Renewable Energy

Climate Change, Resilience and Ecosystem Services, Springer Nature, pp. 267-290 | 2026 | Co-author | DOI: 10.1007/978-3-031-96977-5_16

20. Mapping Climate-Driven Persistent Flooding Using Google Earth Engine and Its Impacts in the Indo-Gangetic Basin

Climate Change and Disadvantaged Communities, pp. 330-348 | 2025 | Co-author | DOI: 10.1201/9781003620747-22

21. Climate Change and Phytoplankton Habitat Suitability in Nainital Lake

Climate Change and Disadvantaged Communities, pp. 315-329 | 2025 | First author | DOI: 10.1201/9781003620747-21

22. GIS-Based Landslide Susceptibility Mapping Using Fuzzy AHP - Multicriteria Evaluation Method for Rudraprayag District, Uttarakhand

Landslide Risk Assessment and Mitigation in India, Springer Proceedings in Earth and Environmental Sciences, pp. 85-104 | 2024 | Co-author | DOI: 10.1007/978-981-97-5485-4_7

23. Modelling Potential Zones of Gangotri Glacier Using GIS and ML in the Wake of Physico-Climatic Factors

Climate Change, Vulnerabilities and Adaptation, Springer Nature, pp. 245-271 | 2024 | First author | DOI: 10.1007/978-3-031-49642-4_14

24. Assessing the Possible Causes of Water Quantity Depletion and Assessment of Water Quality of Naini Lake, Nainital

Climate Change: Issues and Challenges, K S Publishing House, pp. 186-192 | 2020 | First author

Preprints / Early-stage Research Outputs

25. HazWatch: A Geospatial Citizen Science Platform Integrating Multi-Hazard Reporting with Data Access and Spatial Analysis in India

Research Square preprint | 2026 | First author | DOI: 10.21203/rs.3.rs-9734483/v1

26. Integrating PhenoCam and AWS Data to Quantify Stage-Specific Climatic Impacts on Rice Phenology

SSRN preprint | 2026 | Co-author | DOI: 10.2139/ssrn.6087391

27. Spatial Time Series Analysis of Warming Patterns in the Kumaun Himalayas: A Future Perspective

Preprint | 2024 | First author

Conference Papers

28. Impact of Land Use/Land Cover Change on Wetland Transformation of Tarai Region of Kumaun Using Geospatial Techniques - A Case Study of Udham Singh Nagar District

Conference paper | 2019 | First author

29. Tracing Dynamics of Snow Cover in Pithoragarh District Using Geo-spatial Techniques

Conference paper | 2019 | First author

RESEARCH IN THE MEDIA

The Times of India - 11 March 2025

"Kumaon Himalayas to face alarming rise in land surface temp by 2030: Study"

National media coverage of research on changing land-surface temperature in the Kumaon Himalayas.

The Times of India - 20 May 2025

"Himalayan river basins face escalating landslide threats, Jhelum most vulnerable, warns study"

Media coverage of research on landslide susceptibility and risk across Himalayan river basins.

SELECTED INTERNATIONAL COLLABORATIONS

- Landslide research collaboration with **Prof. Daisuke Komori, Japan**.
- Predictive arsenic mapping collaboration with researchers at **Jilin University, China**.
- GIS and economic-geography collaboration with researchers at **Ala-Too University, Kyrgyzstan**.

CURRENT RESEARCH DIRECTIONS

- Further development and research evaluation of HazWatch as a geospatial citizen-science and multi-hazard data platform for India.
- Integrated Himalayan flash-flood research combining rainfall, snowmelt, soil saturation, vegetation, land cover, lithology and terrain with hydrological and geospatial modelling.

FIELD & INSTRUMENTATION EXPERIENCE

- Garmin 64S GPS
- Lutron WA-2015 water multiparameter
- Secchi-disk measurements
- Geocoded water-quality sampling
- Phytoplankton sample collection
- Field-to-GIS data integration

LANGUAGES

English, Hindi, French

References available on request.